

COMPUTER FUNDAMENTALS

1. COMPUTER

1.1 Meaning of Computer

The word **Computer** is derived from the Latin word **Computare**, which means "**to calculate**" or "**to compute**."

Initially, computers were developed only to perform mathematical calculations. However, with the advancement of technology, modern computers can perform a wide variety of tasks such as data processing, communication, scientific research, banking, education, entertainment, business management, and artificial intelligence.

Today, a computer is not merely a calculating machine but a **general-purpose electronic device** capable of performing multiple tasks efficiently.

Exam Fact: *Computare* → Latin word → "To Calculate"

1.2 Definition of Computer

Official Definition

A computer is an electronic, programmable device that accepts data as input, processes it according to a set of instructions, stores it if required, and produces meaningful information as output.

Simple Definition

A computer is an electronic machine that receives data, processes it, stores it when necessary, and produces useful information.

Exam Definition

A computer is an electronic programmable machine that works on the IPO (Input–Process–Output) cycle.

1.3 Understanding the Definition

Every word in the definition has a specific meaning.

A. Electronic

A computer is called an **electronic device** because it works using electrical signals flowing through electronic components such as **transistors, integrated circuits (ICs), capacitors, resistors, and microprocessors**.

Unlike mechanical machines, computers perform operations electronically, making them extremely fast and accurate.

Examples

- Desktop Computer
- Laptop
- ATM Machine
- Smartphone
- Smart TV

Remember: A computer is an **electronic device**, not a mechanical device.

B. Device

A **device** is any machine or equipment designed to perform a specific function.

A computer is called a device because it performs tasks assigned by the user.

Examples

Device	Function
ATM	Cash Withdrawal
Calculator	Mathematical Calculations
Laptop	General Computing
Smartphone	Communication & Computing
Printer	Printing Documents

C. Programmable

A computer is **programmable**, meaning it works according to a **program**, which is a set of instructions written in a programming language.

A computer **cannot think, reason, or make decisions independently**. It executes only those instructions that are stored in its memory.

Example

Suppose you open **MS Paint** and click the **Circle Tool**.

The computer draws a circle **not because it understands your intention**, but because the software already contains instructions telling it what to do when that button is clicked.

Similarly:

- Press **Ctrl + C** → Copy command executes.
- Press **Ctrl + V** → Paste command executes.

The computer simply follows the programmed instructions.

Exam Point: A computer has **no intelligence, emotions, or decision-making ability**. It only follows instructions.

D. Input

Input is the raw data or instructions entered into a computer for processing.

Without input, a computer cannot perform any task.

Common Input Devices

- Keyboard
- Mouse
- Scanner
- Webcam
- Microphone
- Touchscreen
- Barcode Reader

Example

When a student types:

Name : Rahul

Marks : 92

using the keyboard, this is called **Input**.

Another example:

When you insert your ATM card and enter the PIN, the PIN is the **input** provided to the ATM.

E. Processing

Processing is the operation in which the computer converts raw data into meaningful information by performing calculations, comparisons, logical operations, sorting, or decision-making according to the instructions stored in memory.

Processing is performed by the **CPU (Central Processing Unit)**.

Example

Input:

45 + 35

Processing:

The CPU performs the addition.

Output:

80

Another example:

A school enters the marks of all students into the computer.

The computer processes those marks to calculate:

- Total Marks
- Percentage
- Grade
- Rank

This entire operation is called **Processing**.

F. Storage

Storage is the process of saving data, programs, and information so that they can be used again in the future.

A computer stores data in memory devices.

Examples

- Hard Disk Drive (HDD)

- Solid State Drive (SSD)
- Pen Drive
- Memory Card
- CD/DVD

Real-life Example

When you save a project in **Microsoft Word**, the file is stored on your computer's storage device. You can open and edit it later without creating it again.

G. Output

Output is the final meaningful result produced after processing the input.

Output is presented to the user through output devices.

Common Output Devices

- Monitor
- Printer
- Speaker
- Projector

Example

Input:

25×4

Processing:

Multiplication by CPU.

Output:

100

Another example:

After booking a railway ticket online, the generated **e-ticket** displayed on the screen is the **output**.

Real-Life Example: ATM Transaction

Step	Operation
Insert ATM Card	Input
Enter PIN	Input
Bank verifies PIN	Processing
Checks Account Balance	Processing
Cash is Dispensed	Output
Transaction is Saved	Storage

This example demonstrates how all the basic functions of a computer work together in a real-life situation.

Father of Computer

Charles Babbage (1791–1871)

Charles Babbage was an English mathematician, inventor, philosopher, and mechanical engineer. He is known as the **Father of the Computer** because he designed the first mechanical computing machines and introduced the concept of a programmable general-purpose computer.

Although he could not complete his machines during his lifetime due to technological and financial limitations, his designs became the foundation of modern computer architecture.

Major Contributions

Difference Engine (1822)

The **Difference Engine** was a mechanical calculating machine designed to compute mathematical tables automatically. It reduced human calculation errors and is considered one of the earliest automatic calculators.

Analytical Engine (1837)

The **Analytical Engine** was a revolutionary design that introduced features found in modern computers, including:

- Input Unit (Punch Cards)
- Processing Unit ("Mill")
- Memory ("Store")
- Output Unit

- Control Mechanism

Because it had all the basic components of a modern computer, it is regarded as the **first design of a general-purpose programmable computer**.

Why is Charles Babbage called the Father of the Computer?

- Designed the Difference Engine (1822).
- Proposed the Analytical Engine (1837).
- Introduced the concept of programmable computing.
- Laid the foundation for modern computer architecture.

Exam Facts

- Father of Computer → Charles Babbage
 - Difference Engine → 1822
 - Analytical Engine → 1837
-

First Computer Programmer (Mother of Programming)

Ada Lovelace (1815–1852)

Ada Lovelace was an English mathematician who collaborated with Charles Babbage on the Analytical Engine.

She wrote the **first algorithm** intended to be executed by a machine, making her the **world's first computer programmer**.

Ada also understood that computers could perform much more than mathematical calculations. She predicted that computers would one day create music, process symbols, and solve complex logical problems—ideas that were far ahead of her time.

Major Contributions

- Wrote the first computer algorithm.
- Explained how the Analytical Engine could be programmed.
- Introduced the concept of software before computers were actually built.

Exam Facts

- First Computer Programmer → Ada Lovelace

2. IPO Cycle (Input → Process → Output)

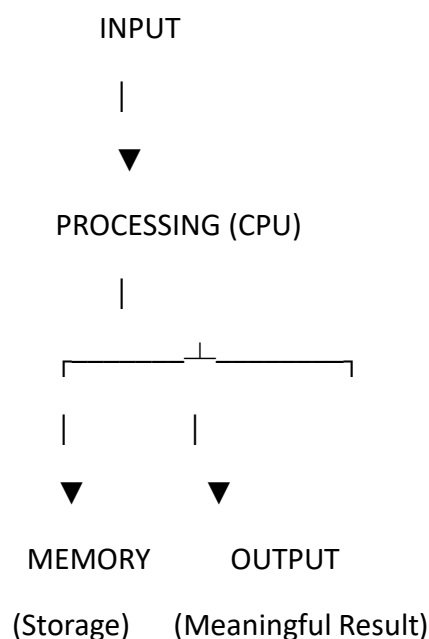
Introduction

Every computer works on a basic operating principle known as the **IPO Cycle (Input–Process–Output Cycle)**. It describes how a computer receives data, processes it according to instructions, stores it when necessary, and produces useful information.

Whether you are using an ATM, mobile phone, calculator, railway reservation system, or laptop, every computing device follows the IPO cycle.

Exam Point: IPO Cycle is the fundamental working principle of a computer.

IPO Flow Diagram



Step 1 : Input

Definition

Input is the raw data, instructions, or commands entered into a computer for processing.

Without input, a computer cannot perform any task.

Common Input Devices

Device	Function
Keyboard	Enter text and numbers

Device	Function
Mouse	Pointing and selection
Scanner	Converts paper documents into digital form
Microphone	Voice Input
Webcam	Image & Video Input
Barcode Reader	Reads barcodes
Touchscreen	Direct touch input
Joystick	Game Control

Examples of Input

- Typing a password.
 - Clicking the Login button.
 - Entering marks into a computer.
 - Uploading a photo.
 - Scanning a QR code.
-

Step 2 : Processing

Definition

Processing is the operation in which the computer manipulates input data according to the instructions stored in memory to produce meaningful information.

Processing is performed by the **CPU (Central Processing Unit)**.

During processing, the CPU performs operations such as:

- Calculation
- Comparison
- Sorting
- Searching
- Decision Making

- Logical Operations
-

Example

Input

25 + 35

↓

CPU performs addition.

↓

Output

60

Another Example

A teacher enters marks of five students.

The CPU calculates:

- Total Marks
- Percentage
- Grade
- Rank

This entire operation is called **Processing**.

Step 3 : Storage

Definition

Storage is the process of saving data, instructions, and information for future use.

The computer stores both the original input and the processed output.

Why is Storage Necessary?

Without storage,

- Files cannot be saved.

- Applications cannot be installed.
 - Photos and videos cannot be kept.
 - The computer would lose all information after shutdown.
-

Storage Devices

- Hard Disk Drive (HDD)
 - Solid State Drive (SSD)
 - Pen Drive
 - Memory Card
 - CD
 - DVD
 - Blu-ray Disc
-

Example

When you press **Ctrl + S** in Microsoft Word, the document is stored on the storage device.

Step 4 : Output

Definition

Output is the meaningful information produced after processing the input data.

Output is presented through output devices.

Common Output Devices

Device	Output Produced
---------------	------------------------

Monitor	Soft Copy
---------	-----------

Printer	Hard Copy
---------	-----------

Speaker	Sound
---------	-------

Projector	Large Screen Display
-----------	----------------------

Device	Output Produced
--------	-----------------

Plotter	Engineering Drawings
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Examples

- ATM displays "Transaction Successful."
 - Railway ticket appears on screen.
 - Calculator displays the answer.
 - Printer prints report card.
 - Speaker plays music.
-

Complete IPO Cycle

Raw Data



Input Devices



CPU Processing



Memory (Storage)



Output Devices



Meaningful Information

Real-Life Examples of IPO Cycle

1. ATM Machine

Step	Operation
Input	Insert ATM Card & Enter PIN
Processing	PIN Verification & Balance Checking
Storage	Transaction Record Saved
Output	Cash Dispensed & Receipt Printed

2. Calculator

Example:

125 + 225

Step	Operation
Input	Enter Numbers
Processing	CPU Performs Addition
Output	350 Displayed

3. Railway Reservation System

Step	Operation
Input	Passenger Details
Processing	Seat Availability Check
Storage	Booking Saved in Database

Step	Operation
------	-----------

Output	E-ticket Generated
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4. Mobile Phone

Step	Operation
------	-----------

Input	Touch Screen / Voice
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Processing Mobile Processor Executes Commands

Storage	Photos & Data Saved
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Output	Screen Display / Speaker Sound
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5. Online Examination

Step	Operation
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Input	Student Answers
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Processing System Evaluates Responses

Storage	Responses Saved
---------	-----------------

Output	Result Generated
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Why is IPO Cycle Important?

- Explains the basic working of every computer.
 - Forms the foundation of computer architecture.
 - Helps understand how data becomes information.
 - Frequently asked in SSC, Railway, Banking, and other competitive examinations.
-

Language Understood by Computer

Introduction

Humans communicate using languages such as **English, Hindi, Bengali, Tamil, and many others**. A computer, however, cannot understand human languages directly. It understands only **Binary Language**, which consists of two digits: **0** and **1**.

Every instruction given to a computer is ultimately converted into binary form before execution.

Exam Point: A computer directly understands only **Machine Language (Binary Language)**.

Binary Language

Definition

Binary Language is the fundamental language of a computer. It uses only **two digits—0 and 1—to represent all data and instructions**.

- **0** represents **OFF** (Low Voltage / False).
- **1** represents **ON** (High Voltage / True).

These two states correspond to the electronic switching inside millions (or billions) of transistors in a processor.

Why Does a Computer Understand Only 0 and 1?

Electronic circuits inside a computer work based on electrical signals.

A transistor can have only two stable states:

Electrical State	Binary Value
OFF (No Current)	0
ON (Current Present)	1

Because hardware naturally distinguishes these two states, binary is reliable, fast, and resistant to electrical noise. For this reason, all data—text, images, videos, music, and programs—is ultimately stored and processed as combinations of **0s and 1s**.

Binary Digit (Bit)

A **Bit (Binary Digit)** is the **smallest unit of data** in a computer.

A bit can hold only one value:

- 0

- 1

Examples:

0

1

Byte

A **Byte** is a group of **8 bits**.

1 Byte = 8 Bits

Example:

01000001

This 8-bit binary pattern represents the letter **A** in the ASCII character encoding system.

Common Memory Units

Unit	Value
1 Bit	0 or 1
1 Nibble	4 Bits
1 Byte	8 Bits
1 KB	1024 Bytes
1 MB	1024 KB
1 GB	1024 MB
1 TB	1024 GB
1 PB	1024 TB

SSC Note: Competitive exams generally follow the binary convention (1 KB = 1024 Bytes), although storage manufacturers often use decimal values for product capacity.

Machine Language

Definition

Machine Language is the lowest-level programming language consisting entirely of binary digits (0 and 1). It is the **only language that the CPU can execute directly** without translation.

Advantages

- Fastest execution.
- No translator required.

Disadvantages

- Very difficult for humans to read and write.
 - Error-prone.
 - Time-consuming to develop programs.
-

Why Do Humans Need Programming Languages?

Writing programs directly in binary is extremely difficult.

Instead, programmers use **high-level programming languages** such as:

- C
- C++
- Java
- Python
- JavaScript

A **compiler** or **interpreter** translates these languages into machine language so the CPU can execute them.

Example

Human-readable instruction:

```
print("Hello")
```

After compilation or interpretation, the processor executes the equivalent binary instructions—not the English words.

Key Differences

Feature	Human Language	Machine Language
Used By	Humans	Computers
Symbols	Alphabets & Words 0 and 1	
Easy to Read	Yes	No
Needs Translation	No	Directly Executed by CPU

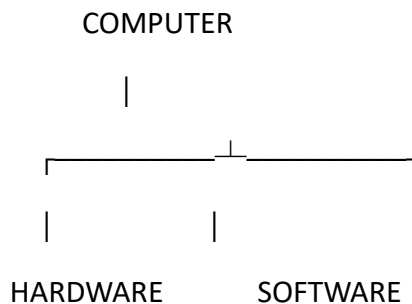
- Also known as → Mother of Programming
- Worked with → Charles Babbage

3. Components of a Computer

Introduction

A computer is not a single machine but a combination of different components that work together to perform a task. These components are broadly classified into **Hardware** and **Software**.

Neither hardware nor software can work independently. Hardware requires software to perform tasks, and software requires hardware to execute instructions.



Exam Point: A computer system consists of **Hardware + Software**.

Hardware

Definition

Hardware refers to the **physical or tangible parts** of a computer that can be **seen, touched, and physically handled**.

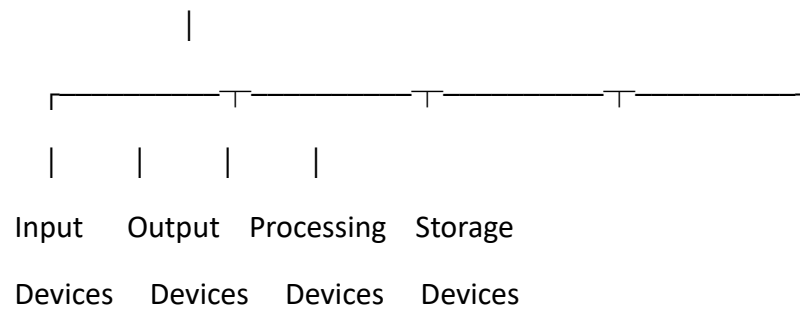
Hardware performs tasks according to the instructions provided by software.

Characteristics of Hardware

- Physical components of a computer.
 - Can be seen and touched.
 - Manufactured using electronic and mechanical parts.
 - Subject to wear and tear over time.
 - Requires software to function.
-

Categories of Hardware

Hardware



A. Input Devices

Definition

An **Input Device** is a hardware device used to enter **data, instructions, or commands** into a computer.

Without input devices, a computer cannot receive information from the user.

Common Input Devices

Device	Function
Keyboard	Enter text, numbers, and symbols
Mouse	Pointing, clicking, selecting
Scanner	Converts paper documents into digital form
Microphone	Voice input
Webcam	Captures images and videos
Joystick	Controls games and simulations
Touchscreen	Input through touch
Barcode Reader	Reads barcode information
OCR Reader	Reads printed characters
OMR Reader	Reads shaded answer sheets
MICR Reader	Reads magnetic characters on bank cheques
Biometric Scanner	Fingerprint/Face Recognition

SSC Note: OCR, OMR, MICR, Barcode Reader, and Biometric Scanner are among the most frequently asked input devices in competitive exams.

B. Output Devices

Definition

An **Output Device** is a hardware device that presents the **processed information** to the user.

Common Output Devices

Device	Output Produced
Monitor	Displays text, images, and videos
Printer	Produces hard copy
Speaker	Produces sound
Headphones	Personal audio output
Projector	Displays output on a large screen
Plotter	Prints engineering and architectural drawings

Soft Copy vs Hard Copy

Soft Copy	Hard Copy
Electronic form	Printed form
Viewed on screen	Viewed on paper
Monitor	Printer

C. Processing Devices

Definition

Processing devices perform calculations, logical operations, comparisons, and execute instructions.

The **Central Processing Unit (CPU)** is the primary processing device of a computer.

Processing Hardware

- CPU
- Motherboard
- Graphics Processing Unit (GPU) (*specialized processor for graphics*)

CPU

The **CPU (Central Processing Unit)** is the **main processing unit** of the computer.

It performs:

- Arithmetic Calculations
- Logical Decisions
- Data Processing
- Program Execution
- Device Control

Exam Point: CPU is known as the **Brain of the Computer**.

(CPU will be discussed in detail in a later section.)

D. Storage Devices

Definition

Storage devices are hardware components used to **store data, programs, and information** permanently or temporarily.

Examples

Device	Storage Type
Hard Disk Drive (HDD)	Magnetic Storage
Solid State Drive (SSD)	Flash Storage
Pen Drive	Flash Storage

Device	Storage Type
Memory Card	Flash Storage
CD	Optical Storage
DVD	Optical Storage
Blu-ray Disc	Optical Storage

Hardware Classification Summary

Category	Examples
Input Devices	Keyboard, Mouse, Scanner, Webcam
Output Devices	Monitor, Printer, Speaker
Processing Devices	CPU, GPU
Storage Devices	HDD, SSD, Pen Drive

Software

Definition

Software is a collection of **programs, instructions, and related data** that tells the hardware **what to do and how to do it**.

Unlike hardware, software **cannot be seen or touched physically**.

Without software, hardware is useless.

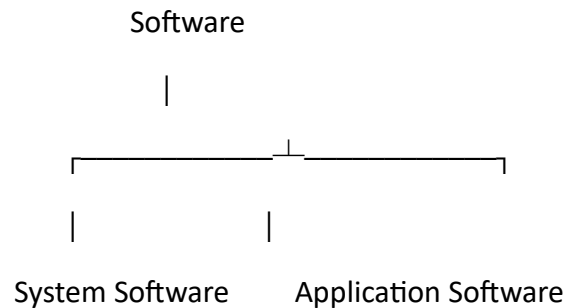
Example: A laptop without an operating system cannot perform useful tasks, even though all its hardware components are present.

Characteristics of Software

- Intangible (cannot be touched).
- Developed using programming languages.
- Controls the operation of hardware.
- Can be installed, updated, or removed.

- Does not wear out physically.

Types of Software



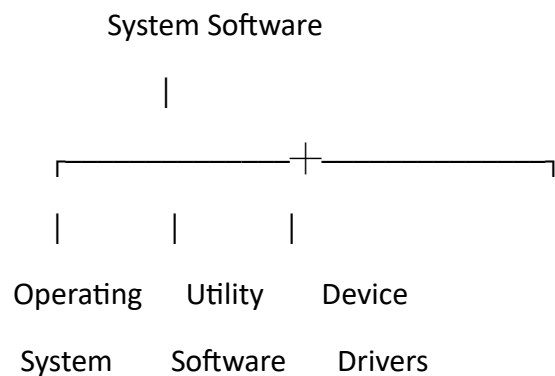
A. System Software

Definition

System Software is software that manages the computer's hardware and provides a platform for running application software.

It acts as an interface between the **user** and the **hardware**.

Types of System Software



Operating System (OS)

Definition

An **Operating System (OS)** is the most important system software that manages all hardware resources and provides an environment for running applications.

Functions

- Memory Management
- File Management
- Device Management
- Process Management
- User Interface
- Security Management

Examples

- Microsoft Windows
- Linux
- macOS
- Android
- iOS

SSC Note: Windows is the most commonly used desktop operating system.

Utility Software

Definition

Utility software helps maintain, protect, and optimize the computer system.

Examples

- Antivirus Software
 - Disk Cleanup
 - Backup Software
 - Disk Defragmenter (*mainly for HDDs*)
 - File Compression Tools
-

Device Drivers

Definition

A **Device Driver** is software that enables the operating system to communicate with hardware devices.

Examples

- Printer Driver
- Graphics Driver
- Sound Driver
- Wi-Fi Driver
- Bluetooth Driver

Without the correct driver, the operating system may not recognize or use the hardware properly.

B. Application Software

Definition

Application software is designed to perform **specific tasks for the user**.

Unlike system software, it focuses on user-oriented activities.

Examples

Software	Purpose
MS Word	Word Processing
MS Excel	Spreadsheet
MS PowerPoint	Presentation
Google Chrome	Web Browsing
VLC Media Player	Audio & Video Playback
Adobe Reader	Reading PDF Files
Photoshop	Image Editing
Tally	Accounting

Hardware vs Software

Hardware

Physical components

Can be touched

Manufactured

Subject to physical damage

Requires software to operate

Example: Keyboard, Monitor, CPU

Software

Programs and instructions

Cannot be touched

Developed by programmers

Can be corrupted or deleted but not physically damaged

Requires hardware to run

Example: Windows, MS Word, Chrome

Relationship Between Hardware and Software

Hardware and software are interdependent.

- Hardware without software cannot perform meaningful tasks.
- Software cannot function without hardware.

Example:

MS Word (software) cannot run without a computer (hardware), and a computer cannot create documents without software like MS Word.

4. Memory

Introduction

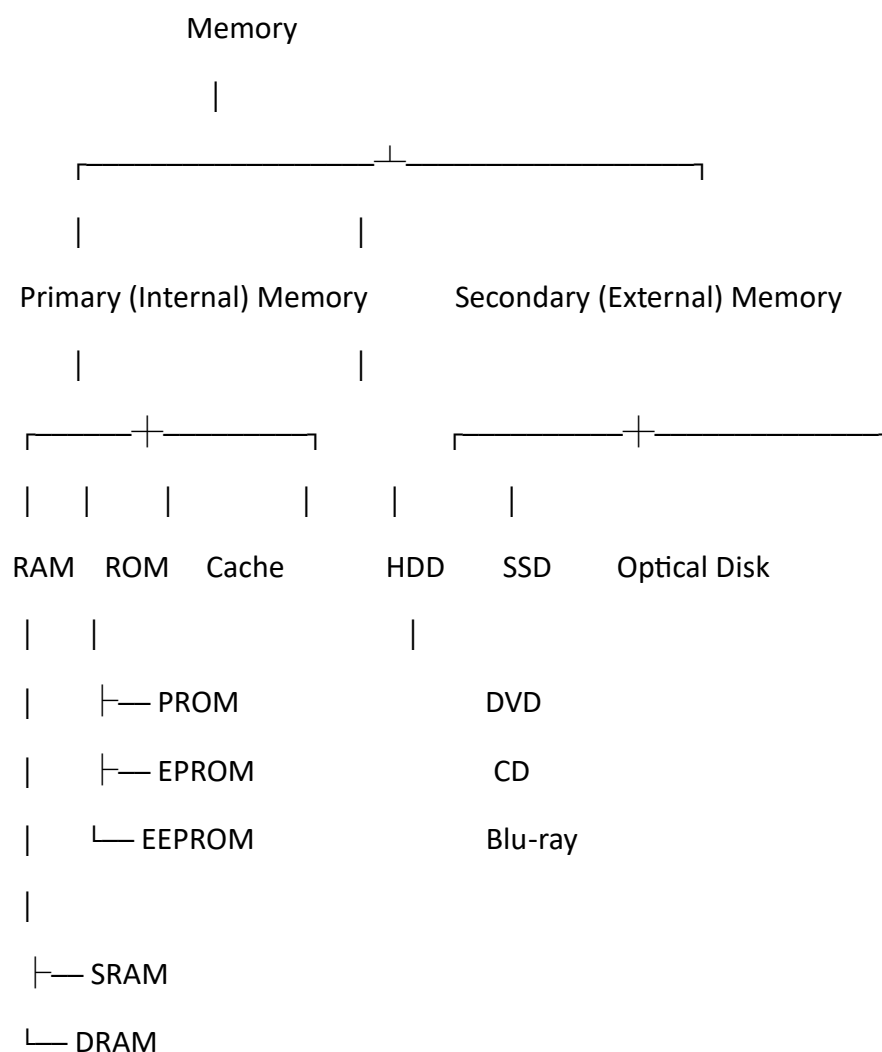
A computer works with enormous amounts of data every second. To process, store, and retrieve this data efficiently, it uses **Memory**.

Memory is one of the most important components of a computer because it stores **data, instructions, programs, and processing results**.

Without memory, a computer cannot execute programs or retain information.

Definition: Memory is the storage unit of a computer that stores data, instructions, programs, and information either temporarily or permanently.

Classification of Memory



Types of Memory

Memory is broadly classified into two types:

1. **Primary Memory (Main Memory / Internal Memory)**

2. Secondary Memory (Auxiliary Memory / External Memory)

Primary Memory (Main Memory)

Definition

Primary Memory is the memory that is **directly accessed by the CPU**. It stores the data and instructions that are currently being processed.

It is called **Main Memory** because the processor communicates with it directly.

Characteristics

- Directly connected to CPU.
 - Faster than Secondary Memory.
 - Smaller storage capacity.
 - Expensive.
 - Required for system operation.
 - Some parts are temporary while others are permanent.
-

Types of Primary Memory

- RAM
 - ROM
 - Cache Memory
-

RAM (Random Access Memory)

Definition

RAM is a **volatile primary memory** used to temporarily store data and programs currently being used by the CPU.

When power is switched OFF, all data stored in RAM is lost.

Definition: RAM is a volatile read-write memory that temporarily stores the data and instructions required by the CPU during processing.

Why is RAM Called Volatile?

Volatile means the memory **loses all stored data when electrical power is removed.**

Example:

You are writing a project in MS Word.

If electricity suddenly goes OFF before saving,

→ Unsaved work stored in RAM disappears.

Characteristics of RAM

- Volatile Memory
- Read and Write Memory
- Temporary Storage
- High Speed
- Directly Accessible by CPU

Uses of RAM

- Running Applications
- Opening Files
- Gaming
- Internet Browsing
- Video Editing
- Multitasking

Types of RAM

RAM

|

|— SRAM

└— DRAM

SRAM (Static Random Access Memory)

Definition

SRAM stores data using **flip-flop circuits**.

It does **not require refreshing** as long as power is supplied.

Characteristics

- Very Fast
 - Expensive
 - Low Storage Capacity
 - Low Density
 - Used as Cache Memory
-

Advantages

- Faster than DRAM
 - Consumes less power during operation
 - No Refresh Cycle Required
-

Disadvantages

- Costly
 - Occupies More Space
 - Smaller Capacity
-

Uses

- CPU Cache
 - High-Speed Buffer
-

DRAM (Dynamic Random Access Memory)

Definition

DRAM stores data in **capacitors**.

The stored charge leaks away with time, so it **must be refreshed thousands of times every second** to retain data.

Characteristics

- Slower than SRAM
 - Cheaper
 - High Density
 - Larger Capacity
 - Requires Refreshing
-

Advantages

- Low Cost
 - High Storage Capacity
 - Used in Main Memory
-

Disadvantages

- Slower
 - Refresh Circuit Needed
-

Uses

- Main Memory of Computers
 - Laptop Memory
 - Desktop Memory
-

SRAM vs DRAM

Feature	SRAM	DRAM
Full Form	Static RAM	Dynamic RAM

Feature	SRAM	DRAM
Storage Method	Flip-Flops	Capacitors
Speed	Faster	Slower
Refresh Required	No	Yes
Cost	High	Low
Capacity	Low	High
Power Consumption	Lower (no refresh)	Higher (refresh needed)
Used In	Cache Memory	Main Memory

SSC PYQ Fact: SRAM is faster but costlier than DRAM.

ROM (Read Only Memory)

Definition

ROM is a **non-volatile primary memory** that permanently stores essential instructions required to start the computer.

The contents of ROM are **not erased when power is switched OFF**.

Characteristics

- Non-Volatile
 - Permanent Memory
 - Mostly Read Only
 - Stores Firmware
 - Used During Booting
-

Uses

- BIOS Firmware
- Embedded Systems
- Electronic Devices

- Boot Program
-

Types of ROM

ROM

|

├— PROM

├— EPROM

└— EEPROM

PROM (Programmable Read Only Memory)

Definition

PROM is manufactured as a blank memory.

The user can write data **only once** using a special device called a **PROM Programmer**.

After programming, the contents **cannot be erased or modified**.

Uses

- Small Electronic Devices
 - Permanent Firmware
-

EPROM (Erasable Programmable Read Only Memory)

Definition

EPROM can be erased by exposing the chip to **Ultraviolet (UV) Light** and then reprogrammed.

Characteristics

- Reusable
- UV Light Required
- Chip Must Be Removed Before Erasing

Uses

- Older Computer Systems
- Industrial Controllers

EEPROM (Electrically Erasable Programmable Read Only Memory)

Definition

EEPROM can be erased and programmed **electrically** without removing the chip from the computer.

It is the most advanced form of ROM among the three.

Characteristics

- Electrically Erased
- Electrically Rewritten
- No UV Light Required
- Can Be Updated Many Times

Uses

- BIOS Chips
- Routers
- Mobile Phones
- Motherboards

Comparison of ROM Types

Feature	PROM	EPROM	EEPROM
Programmable	Once	Multiple Times	Multiple Times
Erasing Method	Cannot Erase	UV Light	Electricity

Feature	PROM	EPROM	EEPROM
Chip Removal Required	No	Yes	No
Reprogrammable	No	Yes	Yes

Memory Trick:

PROM → Program Once

EPROM → Erase using Photons (UV Light)

EEPROM → Electrically Erased

Cache Memory

Definition

Cache Memory is a **very small, extremely fast primary memory** located **inside or very close to the CPU**.

It stores frequently used data and instructions so that the CPU can access them quickly.

Why is Cache Memory Needed?

The CPU works much faster than RAM.

If the CPU had to wait for RAM every time it needed data, system performance would decrease.

Cache Memory reduces this waiting time by keeping frequently used data readily available.

Characteristics

- Fastest Memory in the Computer
 - Very Expensive
 - Small Capacity
 - Located Near/Inside CPU
 - Improves Processing Speed
-

Levels of Cache Memory

CPU

|
├— L1 Cache (Fastest, Smallest)

|
├— L2 Cache (Moderate)

|
└— L3 Cache (Largest, Shared)

L1 Cache (Level 1)

- Located inside CPU Core
 - Fastest Cache
 - Smallest Size (typically tens of KB)
-

L2 Cache (Level 2)

- Larger than L1
 - Slightly Slower
 - Usually dedicated to a CPU core
-

L3 Cache (Level 3)

- Largest Cache
 - Shared among CPU cores
 - Slower than L1 and L2 but much faster than RAM
-

Primary vs Secondary Memory

Feature	Primary Memory	Secondary Memory
Access by CPU	Direct	Indirect
Speed	Faster	Slower
Capacity	Smaller	Larger
Cost	Expensive	Cheaper

Feature	Primary Memory	Secondary Memory
Examples	RAM, ROM, Cache HDD, SSD, DVD, Pen Drive	

Secondary Memory

Definition

Secondary Memory is **non-volatile storage** used to permanently store data, programs, and files.

Unlike primary memory, it is **not directly accessed by the CPU**. Data must first be loaded into RAM before the CPU can process it.

Examples

- Hard Disk Drive (HDD)
- Solid State Drive (SSD)
- CD
- DVD
- Blu-ray Disc
- Pen Drive
- Memory Card

5. Data and Information

Introduction

Every task performed by a computer begins with **Data** and ends with **Information**. A computer itself does not understand whether the input is meaningful or not; it simply processes the data according to the given instructions and produces information.

Understanding the difference between **Data** and **Information** is one of the most fundamental concepts in Computer Fundamentals and is frequently asked in SSC CHSL, RRB NTPC, Banking, and State-level examinations.

Data

Definition

Data refers to **raw, unorganized, and unprocessed facts or figures** that do not have a complete meaning by themselves.

Data can be in the form of:

- Numbers
- Characters
- Symbols
- Images
- Audio
- Video

Until data is processed, it cannot be used for decision-making.

Exam Definition:

Data is a collection of raw facts and figures that has not yet been processed.

Examples of Data

- 95
- Rahul
- Kolkata
- ₹4500
- 25°C
- 12/08/2026

Individually, these values do not provide complete information.

Characteristics of Data

- Raw and unprocessed
 - May or may not have meaning
 - Used as input for processing
 - Can be stored in any format
 - Becomes useful only after processing
-

Information

Definition

Information is **processed, organized, and meaningful data** that can be understood and used for decision-making.

Information is obtained after processing data.

Exam Definition:

Information is processed data that is meaningful and useful for making decisions.

Examples of Information

Data	Information
65, 72, 81, 92	Average Marks = 77.5
₹1200, ₹450, ₹350	Total Expense = ₹2000
Student Marks	Grade Card
Monthly Sales	Profit Report

Characteristics of Information

- Processed data
- Organized
- Meaningful

- Useful for decision-making
- Accurate and understandable

Difference Between Data and Information

Data	Information
Raw facts and figures	Processed data
Unorganized	Organized
Input	Output
Has little or no meaning	Has complete meaning
Cannot directly help in decision-making	Used for decision-making
Example: 95, 82, 76	Example: Average = 84.3

Real-Life Example

Suppose the marks of five students are:

85

76

92

88

79

These are **Data**.

The computer calculates:

- Total = 420
- Average = 84
- Highest = 92

These results are **Information**.

Data Processing

Definition

Data Processing is the process of converting raw data into meaningful information.

During processing, the computer performs operations such as:

- Calculation
 - Sorting
 - Searching
 - Classification
 - Comparison
 - Summarization
-

Data Processing Cycle

Data

|



Processing

|



Information

Example 1 – School

Data

Student Marks:

- English = 82
- Mathematics = 90
- Science = 87



Processing

Computer calculates:

- Total Marks
- Percentage
- Grade

↓

Information

Total = 259

Percentage = 86.33%

Grade = A

Example 2 – ATM

Data

- ATM Card Number
- PIN
- Withdrawal Amount

↓

Processing

- Verify PIN
- Check Balance
- Deduct Amount

↓

Information

"₹5,000 Successfully Withdrawn."

Example 3 – Railway Reservation

Data

- Passenger Name
- Age
- Train Number

- Date



Processing

- Seat Availability Check
- Fare Calculation



Information

Confirmed Ticket Generated

CPU (Central Processing Unit)

Introduction

The **Central Processing Unit (CPU)** is the **most important hardware component** of a computer.

It performs all calculations, logical operations, and controls the functioning of every part of the computer.

Since every instruction is processed by the CPU, it is known as the **Brain of the Computer**.

Definition

CPU (Central Processing Unit) is the primary processing unit of a computer that executes instructions, performs calculations, makes logical decisions, and controls all operations of the computer.

Why is CPU Called the Brain of the Computer?

The CPU is called the **Brain of the Computer** because:

- It controls all hardware devices.
- It executes every instruction.
- It performs arithmetic calculations.
- It makes logical comparisons.
- It manages data flow.

- It coordinates all computer operations.

Just as the human brain controls the body, the CPU controls the computer.

Basic Workflow of a Computer

Input Devices

|



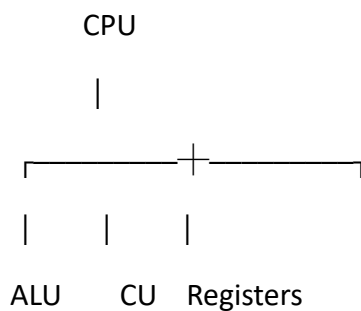
CPU

|



Output Devices

Internal Structure of CPU



The CPU mainly consists of three components:

1. Arithmetic Logic Unit (ALU)
2. Control Unit (CU)
3. Registers

Arithmetic Logic Unit (ALU)

Definition

The **Arithmetic Logic Unit (ALU)** is the part of the CPU responsible for performing **arithmetic operations** and **logical operations**.

Arithmetic Operations

- Addition (+)
 - Subtraction (-)
 - Multiplication (×)
 - Division (÷)
-

Logical Operations

- Greater Than (>)
 - Less Than (<)
 - Equal To (=)
 - Not Equal To (≠)
 - AND
 - OR
 - NOT
-

Example

Input:

15 + 25

↓

ALU calculates

↓

Output:

40

Control Unit (CU)

Definition

The **Control Unit (CU)** is responsible for **controlling and coordinating all operations** of the computer.

It directs the flow of data between the CPU, memory, and input/output devices.

Important: The Control Unit **does not perform calculations**. It only controls and manages operations.

Functions of Control Unit

- Controls execution of instructions.
 - Coordinates input and output devices.
 - Transfers data between memory and CPU.
 - Manages instruction sequence.
-

Registers

Definition

Registers are **very small, extremely fast storage locations** present **inside the CPU**.

They temporarily hold data and instructions currently being processed.

Characteristics

- Fastest memory inside CPU.
 - Very small storage capacity.
 - Temporary storage.
 - Directly accessed by the processor.
-

Uses

- Store intermediate calculation results.
 - Store instruction addresses.
 - Store current processing data.
-

Working of CPU

The CPU performs four basic steps:

Input

|



Fetch Instruction

|



Decode Instruction

|



Execute Instruction

|



Store Result

Step 1 – Fetch

The CPU fetches the instruction from memory.

Step 2 – Decode

The Control Unit interprets the instruction.

Step 3 – Execute

The ALU performs the required operation.

Step 4 – Store

The result is stored in registers or memory and sent to the output device if required.

Clock Speed

Definition

Clock Speed is the rate at which the CPU executes instructions.

It is measured in **Hertz (Hz)**.

Common units:

- Megahertz (MHz)

- Gigahertz (GHz)

Example:

A processor with a clock speed of **3.5 GHz** can execute approximately **3.5 billion clock cycles per second** (actual instructions per second depend on processor architecture).

Computer Speed

Older computer textbooks often state that:

A computer can perform **3–4 million instructions per second (MIPS)**.

This statement refers to older computer systems and is still found in many competitive exam books.

Modern processors, however, can execute **billions of operations per second**, depending on factors such as clock speed, number of cores, cache size, and processor architecture.

Time Units Used in Computing

Unit	Value
1 Millisecond (ms)	10^{-3} second
1 Microsecond (μ s)	10^{-6} second
1 Nanosecond (ns)	10^{-9} second
1 Picosecond (ps)	10^{-12} second

BIOS (Basic Input/Output System)

Introduction

When a computer is switched ON, the operating system (Windows, Linux, etc.) does not start immediately. Before the operating system loads, a small program stored permanently on the motherboard checks whether all hardware components are functioning correctly. This program is called the **BIOS**.

BIOS is the first software that runs when a computer starts.

Exam Point: BIOS is firmware stored on the motherboard.

BIOS

Full Form

BIOS = Basic Input/Output System

Definition

BIOS (Basic Input/Output System) is a firmware program permanently stored in a ROM or Flash Memory chip on the motherboard. It initializes hardware components, performs a Power-On Self-Test (POST), and loads the operating system into memory.

Why is BIOS Required?

Without BIOS:

- The computer cannot recognize hardware devices.
- The operating system cannot be loaded.
- The computer will fail to start properly.

BIOS acts as a bridge between the **hardware** and the **operating system** during startup.

Location of BIOS

BIOS is stored in a special memory chip located on the **Motherboard**.

Older computers used **ROM** or **EPROM**, while modern computers store BIOS in **Flash Memory (EEPROM)** so that it can be updated.

Functions of BIOS

1. Performs POST (Power-On Self-Test)

Checks whether the hardware components are working properly.

Examples:

- RAM
- Keyboard
- Processor
- Hard Disk / SSD
- Graphics Card

If any hardware problem is detected, BIOS generates an error message or beep code.

2. Detects Hardware

BIOS detects all hardware connected to the computer.

Example:

- Mouse
 - Keyboard
 - HDD
 - SSD
 - DVD Drive
 - USB Devices
-

3. Loads the Operating System

After successful hardware testing, BIOS searches for a bootable storage device and loads the operating system into RAM.

4. Provides Basic Input and Output Services

BIOS enables communication between the operating system and hardware devices during the startup process.

5. Stores System Configuration

BIOS stores settings such as:

- Date
- Time
- Boot Order
- CPU Settings
- Security Passwords

These settings are maintained by the CMOS battery.

POST (Power-On Self-Test)

Definition

POST (Power-On Self-Test) is a diagnostic test automatically performed by the BIOS immediately after the computer is powered on.

Its purpose is to ensure that all essential hardware components are functioning correctly before the operating system starts.

Hardware Checked During POST

- CPU
 - RAM
 - Keyboard
 - Mouse
 - Graphics Card
 - Storage Devices
 - Motherboard
-

What Happens if POST Fails?

If a hardware problem is detected:

- The computer may display an error message.
- It may produce a series of beep sounds.
- The operating system will not load until the issue is resolved.

CMOS

Full Form

CMOS = Complementary Metal-Oxide Semiconductor

Definition

CMOS is a small memory chip on the motherboard that stores BIOS configuration settings.

It is powered by a small CMOS battery, allowing it to retain settings even when the computer is turned off.

Information Stored in CMOS

- System Date
 - System Time
 - Boot Sequence
 - BIOS Password
 - Hardware Configuration
-

CMOS Battery

The CMOS battery supplies power to the CMOS chip.

If the battery becomes weak or fails:

- Date and time reset automatically.
 - BIOS settings return to default values.
-

UEFI (Unified Extensible Firmware Interface)

Definition

UEFI (Unified Extensible Firmware Interface) is the modern replacement for the traditional BIOS.

It performs the same basic functions as BIOS but provides improved performance, better security, and support for modern hardware.

Advantages of UEFI

- Faster boot process
- Supports hard drives larger than 2 TB
- Graphical user interface (GUI)
- Mouse support
- Secure Boot feature
- Better security and flexibility

BIOS vs UEFI

Feature	BIOS	UEFI
Full Form	Basic Input/Output System	Unified Extensible Firmware Interface
Interface	Text-Based	Graphical (GUI)
Mouse Support	No	Yes
Boot Speed	Slower	Faster
Maximum Disk Support	Up to 2 TB (traditional MBR limitation)	More than 2 TB (using GPT)
Security	Basic	Secure Boot
Modern Standard	Older Systems	Modern Computers

Exam Point: UEFI is the modern replacement for BIOS.

Booting

Definition

Booting is the process of starting a computer by loading the operating system from a storage device into the computer's main memory (RAM).

Without booting, the operating system cannot run.

Booting Process

Power ON

|



BIOS Starts

|



POST Performed

|



Boot Device Detected

|



Operating System Loaded into RAM

|



Computer Ready to Use

Types of Booting

1. Cold Boot (Hard Boot)

Definition

A **Cold Boot** occurs when a computer is started from a completely powered-off state.

Examples

- Pressing the Power Button after the computer has been shut down.
 - Starting a laptop after the battery was removed and reinserted.
-

Characteristics

- Computer starts from OFF state.
 - Complete hardware initialization occurs.
 - Full POST is performed.
-

2. Warm Boot (Soft Boot)

Definition

A **Warm Boot** occurs when the computer is restarted without switching off the power.

Examples

- Clicking **Restart** in Windows.
 - Pressing **Ctrl + Alt + Delete** (on supported systems).
-

Characteristics

- Computer is already ON.
 - Operating system restarts.
 - Usually faster than a Cold Boot because the system is already powered.
-

Cold Boot vs Warm Boot

Feature	Cold Boot	Warm Boot
Starting Condition	Computer is OFF	Computer is ON
Power Supply	Fresh Power On	Restart Without Power Off
POST	Full POST	May perform a reduced or full POST depending on the system
Speed	Slower	Faster
Example	Pressing Power Button	Restart Option

Practical Example of Booting

Suppose you press the power button on your laptop.

1. Electricity reaches the motherboard.
2. BIOS starts automatically.
3. BIOS performs POST.
4. Hardware devices are detected.
5. BIOS finds the bootable SSD or HDD.
6. Windows (or another operating system) is loaded into RAM.
7. The login screen appears.

This entire sequence is called **Booting**.

Advantages and Disadvantages of Computer

Introduction

A computer has become an essential part of modern life. It is used in education, banking, business, hospitals, railways, scientific research, communication, entertainment, defence, and almost every other field.

The popularity of computers is due to their **speed, accuracy, storage capacity, and automation**. However, computers also have certain limitations because they depend entirely on human instructions.

Exam Point: A computer is a powerful electronic device, but it **cannot think or make decisions on its own**.

Advantages of Computer

A computer possesses several characteristics that make it superior to manual methods of work.

1. High Speed

Definition

One of the greatest advantages of a computer is its **very high processing speed**.

A modern computer can perform **millions or even billions of operations per second**, depending on its processor and architecture.

Example

A bank computer can process thousands of transactions within a few seconds.

A railway reservation system can check seat availability almost instantly.

Exam Fact

- Computer performs calculations much faster than humans.
 - Speed is measured in **MHz** or **GHz** (processor clock speed).
-

2. Accuracy

Definition

A computer produces **highly accurate results** provided that the input data and program are correct.

A computer does not make calculation mistakes due to tiredness or carelessness.

Example

If you calculate:

$$45897 \times 7845$$

A computer gives the correct answer every time.

If the answer is wrong, it is usually because of:

- Incorrect input (**Garbage In**)
- Wrong program
- Hardware or software failure

Important Concept: GIGO (Garbage In, Garbage Out)

If incorrect data is entered, the computer will produce incorrect results because it processes exactly what it is given.

3. Diligence

Definition

Diligence means the ability to perform repetitive tasks continuously **without becoming tired, bored, or losing concentration**.

Unlike humans, a computer can work for long hours with the same speed and accuracy.

Example

A server can process online transactions 24 hours a day without fatigue.

4. Automation

Definition

Once a program and data are provided, the computer performs the required tasks automatically without continuous human intervention.

Example

After scheduling a backup at midnight, the computer automatically performs it every day.

Automatic payroll processing is another example.

5. Large Storage Capacity

Definition

A computer can store enormous amounts of data in a very small physical space.

It can store:

- Documents
- Photos
- Videos
- Software
- Databases
- Audio files

Modern storage devices have capacities ranging from **GB** to **TB**, and even **PB** in enterprise systems.

Example

A 1 TB hard disk can store hundreds of thousands of documents and thousands of videos.

6. Versatility

Definition

A computer is **versatile**, meaning it can perform many different kinds of tasks.

The same computer can be used for:

- Studying
- Gaming
- Programming
- Video Editing
- Banking
- Internet Browsing
- Online Classes
- Graphic Designing

7. Reliability

Definition

A computer consistently produces the same output for the same input under the same conditions.

This makes computers reliable for scientific, financial, and business applications.

8. Multitasking

Definition

A computer can perform multiple tasks simultaneously.

Example

You can:

- Listen to music
- Download files
- Edit a document
- Browse the Internet

at the same time.

9. Communication

Computers make communication fast and easy through:

- Email
 - Video Conferencing
 - Social Media
 - Instant Messaging
 - Cloud Services
-

10. Cost Efficiency

Although purchasing a computer may involve an initial expense, it reduces long-term operational costs by:

- Saving time
 - Reducing paperwork
 - Improving productivity
 - Automating repetitive tasks
-

Summary of Advantages

Advantage	Explanation
High Speed	Performs millions/billions of operations quickly
Accuracy	Produces correct results when given correct input
Diligence	Never gets tired or bored
Automation	Works automatically after instructions
Storage	Stores huge amounts of data
Versatility	Performs many different tasks
Reliability	Produces consistent results
Multitasking	Runs multiple programs together
Communication	Enables fast global communication
Cost Efficiency	Saves time and increases productivity

Disadvantages of Computer

Although computers are extremely useful, they also have several limitations.

1. No Intelligence (IQ)

Definition

A computer **cannot think, reason, or make independent decisions** like a human.

It only follows the instructions provided by the program.

Example

If wrong instructions are given, the computer will still execute them.

Exam Point: A computer has **high processing power but no intelligence (IQ)**.

2. No Feelings or Emotions

A computer has no emotions.

It cannot:

- Feel happiness
- Feel sadness
- Show sympathy
- Understand human emotions

It simply processes data according to programmed instructions.

3. Dependent on Human Instructions

A computer cannot start work by itself.

It requires:

- A program
- Input data
- Human instructions

Without these, it remains idle.

4. Garbage In, Garbage Out (GIGO)

If incorrect data is entered, the output will also be incorrect.

The computer cannot determine whether the input is logically correct.

Example

If a student's marks are entered as **950** instead of **95**, the computer will calculate the result based on **950**, producing an incorrect report.

5. Security Risks

Computers are vulnerable to:

- Viruses
- Malware
- Ransomware
- Hacking
- Data Theft

Proper cybersecurity measures are required to protect data.

6. Health Problems

Excessive computer usage may lead to:

- Eye strain
- Neck pain
- Back pain
- Wrist pain
- Mental stress due to prolonged screen time

Maintaining good posture and taking regular breaks can reduce these risks.

7. Job Displacement

Automation has reduced the need for manual work in some sectors.

Routine jobs in manufacturing, data entry, and other repetitive fields may be affected by computerization.

However, it has also created new jobs in software development, cybersecurity, data analysis, artificial intelligence, and many other fields.

8. Requires Electricity

Most computers require a continuous power supply.

Without electricity (or battery power in portable devices), a computer cannot function.

Advantages vs Disadvantages

Advantages Disadvantages

High Speed	No Intelligence
Accuracy	No Emotions
Diligence	Depends on Instructions
Automation	GIGO Principle
Large Storage	Security Threats
Versatility	Health Issues
Reliability	Requires Electricity
Multitasking	Job Displacement in Some Areas

Memory Tricks (SSC Friendly)

Advantages → "SAD AMV RCC"

S → Speed
A → Accuracy
D → Diligence
A → Automation
M → Memory (Storage)
V → Versatility
R → Reliability
C → Communication
C → Cost Efficiency

Disadvantages → "ING SHJ"

I → No Intelligence
N → No Emotions (*No feelings*)
G → Garbage In, Garbage Out (GIGO)
S → Security Risks
H → Health Problems
J → Job Displacement

- **Computer never gets tired → Diligence**
- **Computer cannot think or take decisions → No Intelligence (No IQ)**
- **Computer has no feelings or emotions**
- **Wrong Input → Wrong Output → GIGO (Garbage In, Garbage Out)**
- **High Speed, Accuracy, Storage, and Automation** are the most commonly asked characteristics of computers.
- A computer is **dependent on programs and user instructions**.
- Computers are powerful tools but **do not possess common sense or independent reasoning**.